

than otherwise similar options with longer times to expiry. As an extreme example, consider a European put option on an underlying that is at zero. This is as good as life gets for the owner of a put option. In this case, the option holder wants the option to expire as quickly as possible so that he can get the strike price X as soon as possible. The sooner expiry, the higher the present value of X that the put holder will receive.⁷

- What about increasing the risk-free rate? Why does that increase the value of calls but decrease the value of puts? Let's just go with the flow and accept this for now. We will return to this concept in a bit.

A.2 Boundaries on Option Prices

Before we discuss the pricing of options, let's start by discussing certain restrictions that option prices must obey in order to avoid arbitrage. It is important to note that the restrictions that follow do not depend whatsoever on any distributional assumptions on the underlying asset.

An American option must be worth at least as much as its European counterpart:⁸

$$\begin{aligned} c &\leq C \\ p &\leq P. \end{aligned} \tag{A.1}$$

An American option lets us do everything that a European option does—and more. We could always choose to never exercise our American option early and thereby mimic the payoff pattern of the European. Thus, Americans must be worth at least as much as Europeans. To be a bit more formal, suppose that the European was worth more than the American. Then we could buy the American and sell the European today, generating positive cash flow. There are two states of nature we need to consider in the future: one in which the European is exercised and one in which it is not. If the European is exercised at time T (the only time at which it may be exercised), then exercise the American at that time. The proceeds from the American will exactly offset our short, resulting in zero cash flows. If the European is not exercised against us, then we are guaranteed nonnegative cash flows in this

⁷As a second example, consider a European call option on a stock that expires in a month. If the company is planning to issue a liquidating dividend, say, the day after expiry, this option is probably going to be worth more than an otherwise similar European call option that expires in, say, two months.

⁸It is implicitly understood that both options are written on the same underlying, have the same strike, and have the same time to expiration T .

state of the world. This would imply arbitrage, and thus the European can be worth no more than the American.⁹

It is important to note that it is sometimes optimal to exercise an American put option early (i.e., prior to its expiration), which means that the second inequality in Equations A.1 may hold with equality. We'll talk more about this later.

The next thing to note for European and American call options is

$$\begin{aligned} 0 &\leq c \leq S \\ 0 &\leq C \leq S. \end{aligned} \tag{A.2}$$

Call options can't be worth less than zero. Right? If we ever come up with a call option price that is less than zero, we probably messed up somewhere.¹⁰ The other side of the restriction merits a bit more discussion. It says that a call option can never be worth more than the underlying. A call option gives us the right to buy the underlying, but we have pay the strike price for it. Since we have to pay something to get the underlying when we exercise the call, it can't be worth more than underlying itself.¹¹ If we ever found a call option that was worth more than the underlying, we'd sell the call and buy

⁹The technique or strategy just employed to show the arbitrage bounds in Equation A.1 is one that we will employ often, known more formally as proof by contradiction. In order to show that a desired result holds true, we first start out by assuming it as false (e.g., in this case, we assumed that $c > C$) and show that this would lead to arbitrage, which we know cannot happen. Thus we conclude that our original assumption (in this case, that $c > C$) was false, and therefore conclude that the opposite must be true (in this case, that $c \leq C$).

¹⁰One time when I was reviewing a homework problem that asked to solve for the price of a call option, a student insisted that the price was $-\$5$. I said I would gladly buy the call from him and pointed out that his price of $-\$5$ meant that when I bought the call from him, he would pay me $\$5$ today to take it. After a moment or two of further thought, he rescinded his offer.

¹¹What about a call option that has a strike of zero? In this case, owning the call is the same as owning the underlying, and Equations A.2 hold with equality. Intuitively, if we own a call option that has a strike of zero, then we will always exercise the call option, since doing so gives us the underlying without having to pay anything for it, which is the same as just owning the underlying. To be a bit more formal, denote by $c(X = 0)$ a call option with a strike of zero. Let's show that $c(X = 0) \geq S$. Suppose this was not true. That is, suppose $c(X = 0) < S$. Then buy the call with a strike of zero and sell the underlying, which is positive cash flow today. At expiration, let's exercise the call option we are long. Note that since the strike is zero, we will have to pay nothing in order to exercise. Upon exercise, we will obtain the underlying that we will use to close out our short position in the underlying. Thus, we have locked in zero cash flows in the future and positive cash flows today. This would be an arbitrage. Thus, it must be that $c(X = 0) \geq S$. At the same time, we also know from the first line in Equation A.2 that $c(X = 0) \leq S$ (this condition is true for all calls and therefore must be true for a call with a strike of zero). So we have found that $c(X = 0) \geq S$ and that $c(X = 0) \leq S$, from which we conclude that it must be that $c(X = 0) = S$.

the underlying (positive cash flow today), and if the call were ever exercised against us, we'd deliver the underlying and get the strike price (positive cash flow again). If we could follow this strategy, we could make some money. But who's going to take the other side of this trade?

For puts, we have the following restrictions:

$$\begin{aligned} 0 &\leq p \leq X \\ 0 &\leq P \leq X. \end{aligned} \tag{A.3}$$

If a put were worth more than its strike, we could sell the put and lend X (which is positive cash flow today). If the put is ever exercised against us, reclaim the X that we lent (plus some interest) in order to pay the strike, and we'll get the underlying. If the put is not exercised against us, the X that we lent will return Xe^{rT} at expiry.¹² Thus, we would have nonnegative cash flows at all points in the future, which would be an arbitrage from whence the restriction follows.

A Lower Bound on Call Prices and Early Exercise of American Calls

Whereas Equations A.2 and A.3 place some meaningful upper bounds (and trivial lower bounds) on the values of puts and calls, we can place some interesting lower bounds on their values as well. For European calls on a non-dividend paying asset,

$$c \geq \max[0, S - Xe^{-rT}]. \tag{A.4}$$

In order to show that this result is true, consider the strategy in Table A.4 that consists of selling the underlying, buying the call, and buying risk-free bonds with a face of X . Notice that the cash flows at expiry from this strategy are nonnegative. In the state of nature in which $S_T > X$, this strategy results in cash flow of zero. And in the state in which $S_T \leq X$, the quantity $X - S_T$ is nonnegative which means that the cash flow generated in this state is nonnegative. Thus, in order to avoid arbitrage, we must have $S - Xe^{-rT} - c \leq 0$, from which the result follows.¹³

From Equation A.4 and the fact that $C \geq c$ it follows immediately that

$$C \geq \max[0, S - Xe^{-rT}]. \tag{A.5}$$

¹²Throughout this section we will assume that interest is continuously compounded. Thus, an amount X that earns interest at a continuously compounded rate r will grow to Xe^{rT} at time T .

¹³If $S - Xe^{-rT} - c > 0$, then this strategy would result in positive cash flow today and nonnegative cash flows in all states of nature in the future, which would be an arbitrage. Thus, it must be that $S - Xe^{-rT} - c \leq 0$. Since $c \geq S - Xe^{-rT}$ and $c \geq 0$, it follows that $c \geq \max[0, S - Xe^{-rT}]$.

Action	Cash Flow Today	Cash Flow at Expiry	
		$S_T \leq X$	$S_T > X$
Sell Asset	S	$-S_T$	$-S_T$
Buy Call	$-c$	0	$S_T - X$
Buy Bonds	$-Xe^{-rT}$	X	X
Net	$S - Xe^{-rT} - c$	$X - S_T$	0

Table A.4: Arbitrage Strategy that Shows $c \geq S - Xe^{-rT}$

And since $r > 0$, it follows that

$$S - Xe^{-rT} > S - X,$$

and so we can conclude that $C > S - X$. The fact that $C > S - X$ is a pretty important result, because it tells us that (in the absence of dividends) it is never optimal to exercise an American call early. Therefore, an American call is worth exactly the same as a European call on an asset that pays no dividends. What if the asset pays dividends?¹⁴ So suppose we own an American call option on a non-dividend-paying asset, and we are absolutely convinced that the underlying is going to go down precipitously. Should we exercise the American call quickly, take delivery of the underlying asset, and sell it in the market before the underlying goes down?¹⁵

A Lower Bound on Put Prices and Early Exercise of American Puts

There is an analogous restriction to Equation A.4 for European puts on a non-dividend paying asset:

$$p \geq \max[0, Xe^{-rT} - S]. \quad (\text{A.6})$$

In order to show this holds, consider the strategy in Table A.5 consisting of buying the asset, buying the put, and selling risk-free bonds with a face of X . The cash flows at expiry from this strategy are nonnegative, and so

¹⁴In this case, early exercise of an American call option may be optimal. Consider, for example, a company that is about to issue a liquidating dividend. In this situation, the owner of an American call option on the stock of this company may find it in her best interest to exercise early, prior to the liquidating dividend.

¹⁵This would not be our optimal strategy. If we do this, we would get $S - X$, but if we instead sell the call, we will get C , and as we showed, it must be the case that $C > S - X$. So if we don't want the call option, then we should sell it in the market instead of exercising early.

Action	Cash Flow Today	Cash Flow at Expiry	
		$S_T < X$	$S_T \geq X$
Buy Asset	$-S$	S_T	S_T
Buy Put	$-p$	$X - S_T$	0
Sell Bonds	Xe^{-rT}	$-X$	$-X$
Net	$Xe^{-rT} - S - p$	0	$S_T - X$

 Table A.5: Arbitrage Strategy that Shows $p \geq Xe^{-rT} - S$

we must have

$$Xe^{-rT} - S - p \leq 0$$

to avoid arbitrage. Since we also know $p \geq 0$, the result in Equation A.6 follows.

Since $P \geq p$, it follows from Equation A.6 that

$$P \geq \max[0, Xe^{-rT} - S]. \quad (\text{A.7})$$

And since we can always exercise an American put early, it must be worth at least as much as what we can get for immediate exercise, $X - S$. That is,

$$P \geq X - S.$$

So what does the fact that $P \geq X - S$ tell us about early exercise of American puts? Very little, unfortunately. Note that the inequality is weak. If it were a strict inequality, then we would be able to say that an American put option is always worth more than its exercise value. But this isn't the case. Sometimes it turns out that American puts should be exercised early in which case $P = X - S$.

A Further Look at Early Exercise of Options on Non-Dividend Paying Assets

So how come we never exercise American calls on non-dividend-paying assets early, but sometimes we exercise American puts early? Let's look at the difference between the proceeds from early exercise of an American call, $S - X$, and the value of keeping the American call alive, C . This difference represents the value of early exercise of the American call. While we don't (yet) have an exact value for the American call option, we can use a

lower bound on the call that we found in Equation A.5, namely, that $C \geq S - Xe^{-rT}$, to make the following realization:

$$\begin{aligned}
 \text{Value of Early Exercise of American Call} &= (S - X) - C \\
 &\leq (S - X) - (S - Xe^{-rT}) \\
 &= -X(1 - e^{-rT}) \\
 &< 0.
 \end{aligned} \tag{A.8}$$

What is this telling us? We have just found that the value of early exercise of an American call option, as measured by the difference between what we get if we exercise, $S - X$, and what we get if we hold on to the call, C , is always negative. This shows us definitively that early exercise of an American call is a bad thing. The holder of an American call option who exercises early forgoes (among other things) the interest that can be earned on the strike price. That is, if the American call is exercised today, it is necessary to pay the strike of X today. If instead the holder of the American call does not exercise today but decides she will wait until time T to exercise, this implies that she will pay out X at time T . But in order to fund a position of X at time T , she only needs to have Xe^{-rT} on hand today. She can invest this amount today in risk-free bonds and it will grow to an amount equal to X at time T . Thus, by exercising early, she pays out X today, whereas by delaying exercise, she need only outlay Xe^{-rT} . The difference between what she pays out if she exercises today, X , and the amount she needs to spend to buy risk-free bonds that will pay off a face amount of X at time T , Xe^{-rT} , is

$$-X(1 - e^{-rT}),$$

which, as shown in Equation A.8, is a negative number.

Now let's use a lower bound on an American put given in Equation A.7 to look at the value of early exercise of an American put:

$$\begin{aligned}
 \text{Value of Early Exercise of American Put} &= (X - S) - P \\
 &\leq (X - S) - (Xe^{-rT} - S) \\
 &= X(1 - e^{-rT}).
 \end{aligned} \tag{A.9}$$

So what does this tell us? Not much, really. For American puts, the value of early exercise is no greater than $X(1 - e^{-rT})$. But this quantity is a positive number. Although this places an upper bound on the value of early exercise, it does not, however, preclude there being positive value associated with early exercise. If the value of early exercise is positive, then early exercise is a good thing; if it's negative, then it's a bad thing. Since this

Action	Cash Flow Today	Cash Flow at Expiry	
		$S_T < X$	$S_T \geq X$
Buy Put	$-p$	$X - S_T$	0
Buy Asset	$-S$	S_T	S_T
Sell Bonds	Xe^{-rT}	$-X$	$-X$
Sell Call	c	0	$-(S_T - X)$
Net	$c - p - S + Xe^{-rT}$	0	0

Table A.6: Arbitrage Strategy Showing $c = p + S - Xe^{-rT}$, European Put-Call Parity

upper bound is positive, all we can conclude is that sometimes early exercise of American puts may be the right thing to do. In fact, we shouldn't have to think too long in order to come up with a simple example that demonstrates this point.¹⁶

European options are very commonly found in the swaps market, but Americans are somewhat uncommon. However, *Bermudan options* (which can be exercised on a multiple discrete number of dates before expiry) are fairly common. In Chapters 5 and 6 we consider the early exercise of Bermudan options and show by way of example that there may be times that these options are optimally exercised early.

A.3 European Put-Call Parity

European put-call parity is a fundamental result that, while very simple, is a helpful tool. For European puts and calls on the same non-dividend-paying underlying, the same strike, and the same expiry, we have the following relationship:

$$c = p + S - Xe^{-rT}. \quad (\text{A.10})$$

The arbitrage argument that gets us there is shown in Table A.6.

Buying the put and the asset, selling the call, and selling risk-free bonds with a face of X results in a strategy that has zero cash flow at all times in the future. Therefore, it must be that $c - p - S + Xe^{-r(T-t)} = 0$ to avoid arbitrage.

¹⁶What should we do if we own an American put and the value of the underlying asset is zero?

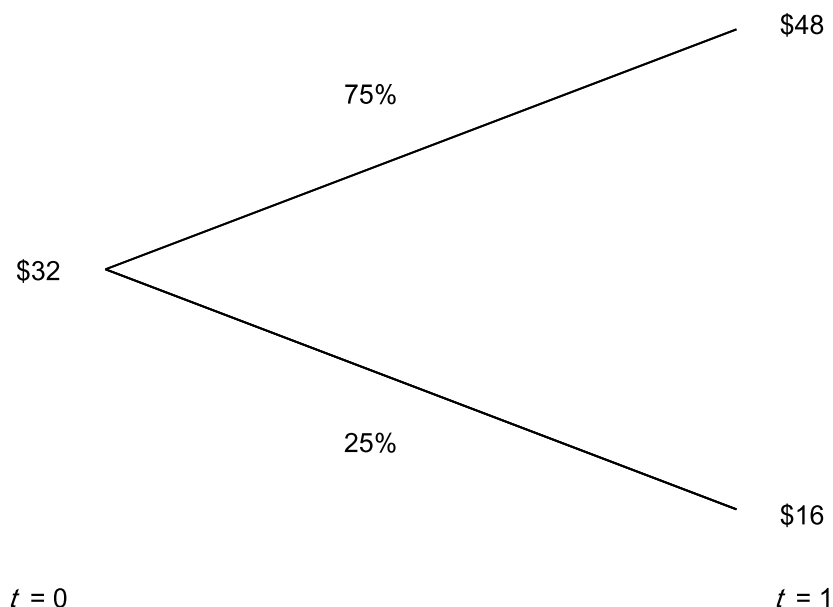


Figure A.1: One-Period Binomial Tree

A.4 Binomial Pricing

So far we haven't really said too much: we defined a bunch of things, showed a few boundary conditions, and showed European put-call parity. That's all fine and well, but the real question is, How much are options worth? Our first crack at tackling this question is by developing the binomial pricing model.

Consider the following example. Figure A.1 depicts an asset that costs \$32 today (time $t = 0$). At time $t = 1$, the asset price will be either \$48 with a 75% probability or \$16 with a 25% probability. How do we know these probabilities? Who really cares? If it makes you happy, pretend we flip a fair coin twice: if it comes up heads twice the asset will be worth \$16; otherwise, it's worth \$48. Also, assume that the (simple) risk-free rate over the period is $33\frac{1}{3}\%$ (in other words, a \$1 investment in the risk-free asset at time $t = 0$ will return $\$1.\overline{33}$ at time $t = 1$).

Now consider a European call option on this asset with an exercise price of \$40 that expires at $t = 1$. Figure A.2 shows the payoff of the call option at time $t = 1$. At the upper node, the call is in-the-money and the payoff is

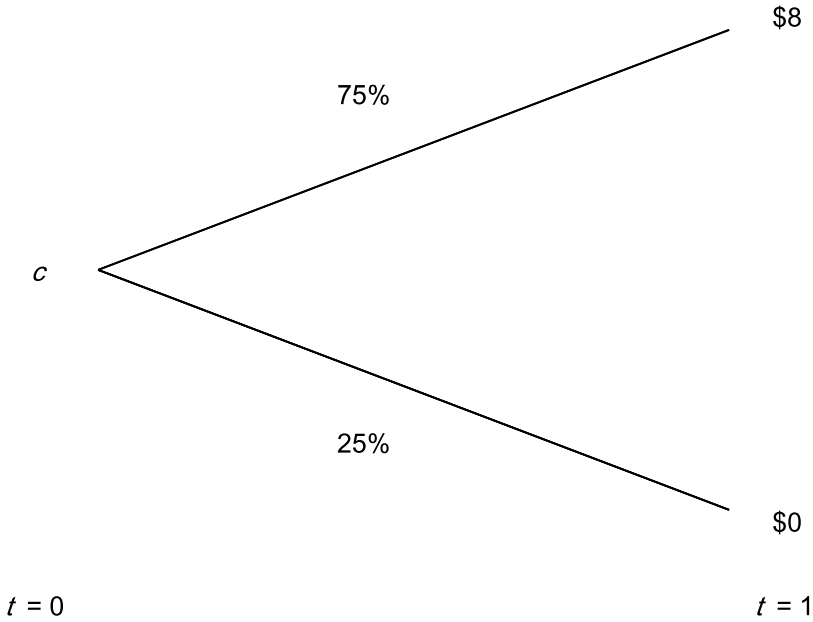


Figure A.2: Payoff of a Call Option

$\max[\$48 - \$40, 0] = \$8$, and at the lower node the call is out-of-the-money and the payoff is $\max[\$16 - \$40, 0] = \$0$. How much is this call worth today at time $t = 0$? Well, that's what we are trying to figure out. First, try to answer the following questions:

- What's the expected terminal value of the asset?
- What's the expected return of the asset?
- What's the expected payoff of the call?

You can look down below at the footnote to see the answers to these three questions.¹⁷ Now that you've answered the above three questions, how about these three:

- What's the price of the call?
- How will investors' risk preferences influence the price of the call?

¹⁷The answers are: $0.75 \times \$48 + 0.25 \times \$16 = \$40$; $(40 - 32)/32 = 25\%$; and $0.75 \times \$8 + 0.25 \times \$0 = \$6$.

- How will the probabilities of an up or down movement affect the price of the call?

I suspect that you might figure that the price of the call is equal to the expected payoff of the call, which is \$6, discounted by the risk-free rate, which is $\$6/1.\overline{33} = \4.50 . This seems straightforward enough.

The other two questions may seem a bit more subtle. The first time I was posed with them, I argued that the more risk-averse investors were, the lower the price of the call, since the call option essentially offers a risky payoff of either \$8 or \$0. And I also thought that the higher the probability that the payoff of the call would be \$8, the greater the value of the call option. Seems reasonable, doesn't it? But it's all wrong, horribly wrong.

To see why, consider a *hedge portfolio* consisting of h units of the underlying asset and borrowing B such that

$$h \times 48 - B \times 1.\overline{33} = 8$$

$$h \times 16 - B \times 1.\overline{33} = 0.$$

We can solve for h and B that satisfy these two equations and find $h = 0.25$ and $B = 3$. Entering into this hedge portfolio requires an initial investment or outlay of cash equal to \$5 (buy 0.25 shares at \$32 a share, which costs \$8, and partially finance the purchase by borrowing \$3).¹⁸ And of course this hedge portfolio was designed so that at $t = 1$ the portfolio could be unwound, and the proceeds would be equal in value to the payoff of the call in either state. To verify this, note that the 0.25 shares will be worth $0.25 \times \$48 = \12 in the upper state or $0.25 \times \$16 = \4 in the lower state, we will have to repay the borrowings plus interest of $\$3(1.\overline{33}) = \4 , and therefore we will be left with either \$8 or \$0 in the upper or lower nodes, respectively.

But this means that the call must be worth \$5 today. By the assumed lack of arbitrage, since we found a hedge portfolio that replicates the payoff of the call, the call option and the hedge portfolio must be worth the same today. Thus, the call is worth \$5, not \$4.50, which was the number we came up with by (incorrectly) discounting the expected payoff of the call by the risk-free rate.

Notice that investor risk preferences never entered into this discussion anywhere. Any investor, regardless of his or her risk preferences, could set up the same hedge portfolio and replicate the call payoffs. That is, the values of h and B do not depend at all on how risk averse an investor is, and therefore option prices do not depend on risk preferences.

In addition, note that the up and down probabilities of 75% and 25% never entered into the discussion. Thus, the call price does not depend on

¹⁸We are making the assumption that one can buy fractional shares of the underlying.